

Hi, I'm

Jinho Park.

AI Application Engineer

I work closely with clients to turn real requirements into AI systems that ship.

AI Application Engineer focused on Information Retrieval and scalable backend systems — vector search, Agentic RAG, latency optimization, and retrieval precision. Currently at Boeing, previously at Lomin.

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About Me

I'm an AI Application Engineer specializing in **Information Retrieval** and **scalable backend systems** — architecting production-grade vector search and Agentic RAG systems with a focus on latency optimization and retrieval precision.

I have a proven track record of automating the end-to-end ML lifecycle using **Prefect**, **Kubernetes**, and **GitOps**, and deep expertise in bridging the gap between complex IR research and high-availability **FastAPI** services.

One thing I'm known for: **fast prototyping**. I'd rather validate an idea with a working demo in days than theorize for weeks — it keeps iteration with clients tight, cheap, and grounded in what actually works.

Currently at **Boeing**, where I work on FAA-compliance RAG and the Boeing AI Factory training framework. Before that, at **Lomin**, I optimized Transformer inference serving with NVIDIA Triton and built document-understanding systems for financial clients.

I graduated from Konkuk University in 2022 with a B.S. in Electrical and Electronics Engineering, and spent a semester on exchange at the University of Agder in Norway.

Technologies I work with

LANGUAGES

Python

C++

RETRIEVAL & VECTOR DBS

Qdrant

Hybrid search

Sparse/Dense embeddings

ML FRAMEWORKS

PyTorch

LangChain

Transformers

MLflow

HARDWARE OPTIMIZATION

NVIDIA Triton

OpenVINO

ONNX Runtime

Quantization

BACKEND & APPLICATION

FastAPI

Redis

PostgreSQL

Streamlit

INFRASTRUCTURE & AUTOMATION

Prefect

Kubernetes

Docker

Helm

GitLab CI/CD

ArgoCD

Git

Linux

OBSERVABILITY & METRICS

Grafana

Prometheus

KPI tracking

SPOKEN LANGUAGES

Korean · Native

English · Advanced

Norwegian · Beginner

Experience

AI Application Engineer · [Boeing](#) ↗

Jan 2024 – present

ML Team · Seoul, South Korea

Advanced Information Retrieval — *Agentic RAG for aviation regulatory compliance*

- Architected a high-precision **Agentic RAG** system that cross-references technical aviation reports against regulatory frameworks through an agentic loop.
- Built it as a dedicated **retrieval service on Qdrant**, then extended it into a **generation app** consuming the retrieval layer as an internal API.
- Built a **hybrid search** pipeline on **Qdrant** (sparse + dense) from scratch, outperforming a dense-only configuration on MRR across a 200-query evaluation set for domain-specific technical terminology.
- Tuned Qdrant indexing and payload filtering to hold **p95 retrieval latency under 1000 ms on CPU-only infrastructure**.
- **CPU-only embedding serving**. The embedding stage ran on CPU only (the downstream internal LLM ran on GPU, managed by a separate team). I sustained throughput with **BentoML** dynamic batching + **ONNX Runtime / OpenVINO** + quantization on CPU.
- Stood up a **Prometheus & Grafana** monitoring stack for the retrieval service, plus a custom **Postgres-backed KPI tracker** (CTR, query quality, adoption, latency SLOs) visualized in Grafana.
- Delivered a user-friendly interface with **Streamlit** and **React** for compliance owners generating regulator-ready reports.

LLM Applications & Internal Tooling

- Built a **GitLab merge-request bot** that lets engineers chat with the internal LLM directly on MRs — code review, summarization, and Q&A inside the review flow.
- Shipped an **MCP (Model Context Protocol) server** exposing internal tools and data sources to LLM agents through a standard interface.
- Built a **complex PDF parser** that extracts document layout via few-shot prompting over (v)LLM APIs; it now powers the data-ingestion pipeline for the compliance RAG system and other internal tools.

MLOps & Backend Automation

- Implemented a **GitOps** deployment flow with **ArgoCD** and **OpenShift**, reducing deployment cycles from **1 hour to 10 minutes**.
- Developed a fault-tolerant **Prefect** orchestration layer, automating data ingestion and inference pipelines for aircraft predictive maintenance.
- Eliminated **~8 hours of manual weekly labor** by building an automated PDF-generation backend that converts raw telemetry into stakeholder-ready reports via FastAPI.

Boeing AI Factory

- Developed a scalable deep-learning training framework enabling efficient AI model development across classical ML, deep learning, and RL.
- Reduced redundant model-development effort by designing a reusable framework for both AI and non-AI engineers.
- Built on **Ray & PyTorch Lightning**; integrated with Boeing's MLOps stack (ClearML, model hub, dataset hub) for seamless deployment.

Machine Learning Engineer · [Lomin](#) ↗

May 2022 – Aug 2023

ML Team · Seoul, South Korea

Inference & Serving Optimization

- Architected a high-throughput serving layer using **NVIDIA Triton**, implementing KV-caching and quantization that accelerated Transformer inference by **20%**.
- Engineered a multi-modal **Table Extraction** system for financial documents, improving **F1 by 15%** and deploying it for enterprise clients.

Retrieval & Fine-tuning

- Developed a **Document Understanding (DU)** framework, building specialized data pipelines for graph-based augmentation and efficient dataset curation.

Clients & Projects

- **Key clients:** Delivered ML solutions for Kyobo Life, Lina Life, Shinhan Asset Trust, and KIPO.
- **PoC initiatives:** Led ML-driven PoC projects for Shinhan Asset Trust, Samsung Fire & Marine Insurance, Hanwha Life, Samsung Life, and KB Insurance.

Software Engineer Intern — OpenVINO Korea · [Intel](#) ↗

Aug 2021 – Feb 2022

OpenVINO Korea · Seoul, South Korea

- Collaborated with local and international teams to analyze the performance of deep learning models on Intel hardware.
- Optimized [OpenVINO](#) components for Intel integrated GPUs, resolving **5+ major performance issues** to enhance inference efficiency.

Education

B.S., Electrical and Electronics Engineering · [Konkuk University](#) ↗

Mar 2015 – Feb 2022

- Studied embedded systems and semiconductors along with low-level programming languages.
- **Thesis:** *Further Optimize MobileNetV2 with Channel-wise Squeeze and Excitation.*
- **CS courses:** Algorithms, Data Structures, Operating Systems, Embedded Systems, Artificial Intelligence.

Exchange Student, Computer Science · [University of Agder](#) ↗

Aug 2017 – Jan 2018

- Studied application development, programming language concepts, and Norwegian.
- Gained international exposure, expanded professional networks, and improved English communication skills.

Achievements & Certifications

Silver Medal — UBC Ovarian Cancer Subtype Classification · [Kaggle](#) ↗ Jan 2024

Ranked **28 of 1,724** participants (top 2%) in the UBC Ovarian Cancer Subtype Classification and Outlier Detection (UBC-OCEAN) competition.

2nd Prize — College Student Papers & Capstone Design Contest · Korean Institute of Broadcast and Media Engineers Nov 2021

Paper: "[Further Optimize MobileNetV2 with Channel-wise Squeeze and Excitation.](#)"

Certified Kubernetes Administrator (CKA) · [Linux Foundation](#) ↗ Dec 2024

Build a Modern Computer from First Principles: Nand to Tetris · [Coursera](#) ↗ Jan 2021

Deep Learning Specialization · [Coursera](#) · [deeplearning.ai](#) ↗ May 2019

Machine Learning · [Coursera](#) · [Stanford](#) ↗ Feb 2019

Open Source

Add BROS model to 🤗 Transformers · [huggingface/transformers#23190](#) ↗ Sep 2023

Added the **BROS** (BERT Relying On Spatiality) document-understanding model to the [Hugging Face Transformers](#) library.

- Ported the model architecture, tokenization, and pre-processing for spatially-aware document inputs.

- Shipped with docs, model card, and integration tests.

[GPU] Fix issue of `prepare_output` —

Oct 2021

OpenVINO · [openvinotoolkit/openvino#7783](https://github.com/openvinotoolkit/openvino/issues/7783) ↗

Fixed a GPU-path issue in `prepare_output` during my Software Engineer Internship at Intel, strengthening correctness for Intel-integrated-GPU inference.

Contact

My inbox is always open. Whether you want to talk about retrieval systems, ML infra, or a role — I'll do my best to reply.

Email me